# TWO-SAMPLE T-TEST

We wish to use a two-sample t-test to determine whether there is any real difference in mean sepal width for the **Versicolor** and **Virginica** species of iris.

We can assume that sepal width is normally distributed, and that the variance for the two groups is equal.

The null hypothesis: there is no real difference in mean sepal width for the two species The alternative hypothesis: there is a difference.

```
head(iris,10)
```

```
##    Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1           5.1         3.5          1.4         0.2  setosa
## 2           4.9         3.0          1.4         0.2  setosa
## 3           4.7         3.2          1.3         0.2  setosa
## 4           4.6         3.1          1.5         0.2  setosa
## 5           5.0         3.6          1.4         0.2  setosa
## 6           5.4         3.9          1.7         0.4  setosa
## 7           4.6         3.4          1.4         0.3  setosa
## 8           5.0         3.4          1.5         0.2  setosa
## 9           4.4         2.9          1.4         0.2  setosa
## 10          4.9         3.1          1.5         0.1  setosa
```

```
t.test(Sepal.Width ~ Species,
       iris,
       Species %in% c("versicolor", "virginica"),
       var.equal = T)
```

```
##
##  Two Sample t-test
##
## data:  Sepal.Width by Species
## t = -3.2058, df = 98, p-value = 0.001819
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.33028246 -0.07771754
## sample estimates:
## mean in group versicolor  mean in group virginica
##                    2.770                    2.974
```

The 95% confidence interval for the difference is –0.33 to –0.08, meaning that the mean sepal width for the **Versicolor** species is estimated to be between 0.08 and 0.33 cm less than for the **Virginica** species.

The p-value of 0.001819 is less than the significance level of 0.05, so we can reject the null hypothesis that the mean sepal width is the same for the **Versicolor** and **Virginica** species in favor of the alternative hypothesis that the mean sepal width is different for the two species.